

Lab #5 - Gaussian Beams

ECE 482/582, Fall 2003

Goals:

1. Verify that the transverse beam intensity for a laser operating in the fundamental or TEM₀₀ mode follows a Gaussian profile.
2. Calculate and experimentally verify the beam waist and divergence angle for a typical HeNe laser using a commercial beam profile instrument, the Photon, Inc. Beam Scan.
3. Examine and compare the beam shapes of the HeNe laser and a collimated visible diode laser.

Reading:

Kuhn, *Laser Engineering*, Chapter 4: section 4.3 – 4.4

Other Resources:

Two classic papers on the subject – both are posted on the lab web site.

Kogelnik and Li, Laser Beams and Resonators, Proc. IEEE, vol. 54, no.10, pp. 1312-1329, Oct.1966

Kogelnik, On the Propagation of Gaussian Beams of Light Through Lenslike Media Including those with a Loss or Gain Variation, Applies Optics, Vol. 4, no. 12, pp. 1562-1569, Dec.1965

Yariv, Quantum Electronics, 3rd Edition, (Wiley and Sons), Section 6.6

Java applet - www.iap.uni-bonn.de/lehre/ss01_laserphysics/GaussianBeam.html

Preparatory Questions:

1. The laser formerly used in this lab was a Spectra Physics Model 086-3 HeNe laser (632.8 nm). The cavity parameters are: mirror 1 radius of curvature $R_1 = \text{infinity}$, mirror 2 radius of curvature $R_2 = 268 \text{ mm}$, mirror spacing $L = 24 \text{ cm}$. Evaluate w_0 , z_0 , and the far-field divergence angle $\theta_{1/2}$ for this laser. What should the size of the spot be at a distance of 5 m from the output mirror?
2. Spectra Physics sometimes curves the backside of the output mirror with a radius of curvature less than the front side radius of curvature. Sketch the output mirror shape. What effect will this have on the expected beam spread? (Take a look at the JAVA applet for this one)

Laboratory Experiment:

The transverse intensity profile of the TEM₀₀ mode is a Gaussian distribution and is given by

$$I(r) = I_0 e^{\frac{-2r^2}{\omega(z)^2}}$$

where I_0 is the peak beam intensity at $r = 0$, r is the transverse beam radius, n is the refractive index of the medium (usually air so $n = 1$), and $w(z)$ is the spot size at a distance z . The spot size represents the beam radius at which the intensity has decreased to $1/e^{-2}$ times its peak value. The peak value I_0 occurs at $r = 0$. Note that as z approaches infinity, $\omega(z)$ varies as z and hence $I(r)$ goes as $1/z^2$. In the far field then, the beam acts as an expanding spherical wave with a finite angular extent.

There is another parameter appearing in the above equations, which is of particular interest. The beam waist at a distance z is given by:

$$\omega(z) = \omega_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}$$

The waist minimum radius is ω_0 . It is a function of the cavity length of the laser L , the radii of curvature of the mirrors R , and the wavelength of the light.

$$\left(\frac{\pi \omega_0^2 n}{\lambda_0}\right) = \frac{d(R_1 - d)(R_2 - d)(R_1 + R_2 - d)}{(R_1 + R_2 - 2d)^2}$$

The last major beam property we are interested in is the far field divergence angle. This is given by

$$\theta_a = \frac{\lambda_0}{\pi n \omega_0}$$

where $\theta_{1/2}$ is the half angle shown in the Figure 4.3 in the text (Kuhn). The divergence angle can be used to estimate the spot sizes of a focused beam.

Procedure:

1. Make measurements of the HeNe transverse beam intensity at several values of z , (from $z < z_0$ to $z \gg z_0$). This is done by carefully aligning the center of the HeNe laser beam reflecting off the rotating mirror to pass directly across the pinhole in front of the Si PIN detector. Plot the oscilloscope traces and use the oscilloscope

cursors to measure x at several points on the trace. These measurements will be used to calculate the radius r of the beam.

2. Shine the laser beam against a far wall and record the divergence angle. [Note: Measure the beam "diameter" of the visible spot you can easily see. This is a good approximation of the beam waist w .] Use the formula for the far field divergence angle to estimate the minimum beam waist.
3. Use the Photon Inc. "Beam Scan" to record the transverse profile and beam diameter of the HeNe laser with z equal to those chosen for the measurements for part 1.
4. Place the 4" focal length lens in front of the laser to change the beam parameters and re-measure the beam diameter as a function of z for the focused beam. What is the minimum spot size you measure at the focus of the lens? What is the new divergence angle of the beam?

Questions to Answer:

- Make a plot of the $\ln(I/I_0)$ vs. r^2 for one value of z (in the far field) which you measured using the rotating mirror.
- Using your plot in part 1 verify that the beam profile is indeed Gaussian and determine the "spot size" $w(z)$. For the remaining values of z , you can just read off the $1/e^2$ power point from the Beam Scan once you have used a scope to check this value from the Beam Scan unit.
- Knowing $w(z)$, calculate the beam waist and beam divergence for the laser without a lens. Does the minimum beam waist occur at the flat laser mirror, as it should? Repeat the above for the experiments run using the lens of your choice.
- Theoretically verify that the equation given earlier for intensity is correct. Is the signal on the scope from the photodetector a measure of the optical electric field strength or the intensity? Explain.
- Why must the pinhole used on the detector be much smaller than the beam diameter?
- Comment on anything you observed that didn't quite fit the "conventional theory".

Conclusion:

After this lab you should have a good understanding of Gaussian beams and how to calculate important beam parameters. A solid knowledge of Gaussian beams is very important when working with lasers and optical systems that use lasers, such as fiber optic communications, free space communications, LIDAR, LADAR, spectroscopy, etc.